SAHAY

Complete Product Documentation & Technical Blueprint

MASTER PRODUCT & TECHNICAL REFERENCE

Seven connected modules • Household-first design • Privacy by default

Document purpose: Master reference for product understanding, development, architecture, implementation, and future expansion.

1. Introduction to SAHAY

1.1 What is SAHAY?

SAHAY is an AI-powered household life-management platform designed to help Indian families organize, manage, understand, and monitor important aspects of their everyday lives from one centralized application.

A typical family manages many different responsibilities:

- Important identity and financial documents.
- Household bills and recurring payments.
- Insurance policies and renewals.
- Medical records and prescriptions.
- School fees and educational documents.
- Family savings goals.
- Household responsibilities and reminders.
- Emergency information.
- Access to private family information.

These responsibilities are usually distributed across physical files, mobile phones, WhatsApp messages, email accounts, spreadsheets, and different applications.

This creates several problems:

1. Important documents become difficult to locate.
2. Payment deadlines are missed.
3. Insurance policies and documents expire without notice.
4. Family members may not know about important household responsibilities.
5. Medical information may not be available during emergencies.
6. Families struggle to monitor savings goals.
7. Information is fragmented across multiple platforms.
8. Understanding complicated documents requires manual effort.

SAHAY addresses these problems by providing a centralized, secure, AI-powered household management system.

1.2 SAHAY's Main Objective

The primary objective is to make household management easier, more organized, secure, and intelligent.

SAHAY should help families:

- Store important information securely.
- Understand documents automatically.
- Monitor household responsibilities.
- Remember important deadlines.
- Manage family health information.
- Track savings goals.
- Ask questions about their household information using AI.

- Control who can access private family information.

1.3 Product Philosophy

1. **One platform:** Family management should not require multiple disconnected applications.
2. **AI-assisted:** AI should reduce repetitive work and help users understand information.
3. **Privacy-first:** Household information must remain private and protected.
4. **Permission-based:** Every family member should only access information they are authorized to access.
5. **Human-controlled:** Important financial, medical, and privacy-related actions require appropriate user confirmation.
6. **Simple experience:** Users should not need technical knowledge to use SAHAY.
7. **Reliable information:** AI-generated answers should be based on available and verified information.
8. **Indian household context:** The product should support common Indian documents, financial obligations, and family structures.

2. Product Structure

SAHAY consists of exactly seven core modules.

No.	Module	Primary responsibility
1	Family Vault	Securely stores family documents
2	Document Doctor	Reads, understands, and extracts information from documents
3	Sahay Daily	Manages household activities, deadlines, and finances
4	Health Shield	Organizes family health information and emergency details
5	Lakshya Goal Vaults	Manages family savings goals
6	Sahay AI Assistant	Provides conversational intelligence across SAHAY
7	Sahay Shield	Provides security, privacy, permissions, and trust infrastructure

These modules are connected. They should not operate as seven independent applications.

For example: A user uploads an electricity bill into Family Vault. Document Doctor reads the bill and extracts its amount and due date. After verification, the information becomes an obligation in Sahay Daily. Sahay Daily displays the deadline and sends reminders. The AI Assistant can answer questions about the bill, subject to the user's permissions.

2.1 Foundational Infrastructure

In addition to the seven modules, SAHAY requires shared infrastructure:

- User registration and authentication.
- Household creation and membership.
- Role management.
- Database management.
- File storage.
- Notifications.
- Background processing.
- API management.

- AI service integration.

Authentication and household management are foundational infrastructure, not additional core modules.

3. Users and Household Structure

SAHAY is designed around households rather than isolated individual accounts.

3.1 User Types

1. Household Owner

The household owner creates and manages the household.

Responsibilities:

- Create the household.
- Invite family members.
- Manage household membership.
- Assign appropriate permissions.
- Configure shared information.
- Manage household-level settings.
- Review security and access activity.

2. Adult Member

An adult member can manage their own information and participate in shared household activities.

Capabilities depend on permissions:

- Upload personal documents.
- Manage authorized household obligations.
- Manage health information.
- Participate in savings goals.
- Access shared family information.
- Approve or decline access requests concerning their private documents.

An adult's private documents must not automatically become visible to another adult simply because both belong to the same household.

3. Elder Member

An elder member should receive a simplified and accessible experience.

Potential capabilities:

- View authorized documents.
- Access health information.
- View reminders.
- Access emergency information.
- Participate in authorized family activities.

4. Caregiver / Helper

A caregiver or helper can access specific information when authorized.

Examples:

- View a medicine schedule.

- Access an emergency health card.
- View a particular document.
- Help manage a household task.

Caregivers should not automatically receive complete household access.

5. Child Member

A child account should have restricted permissions appropriate to the household's configuration.

Possible capabilities:

- View authorized educational documents.
- View assigned tasks.
- Participate in approved goals.
- Access permitted information.

3.2 Household Membership

A household can contain multiple members.

Each member has:

- User ID.
- Household ID.
- Name.
- Email or phone number.
- Household role.
- Membership status.
- Permission settings.
- Account status.

Membership should support:

- Invitations.
- Invitation acceptance.
- Invitation rejection.
- Member removal.
- Role updates.
- Permission changes.
- Membership history.

3.3 Important Privacy Rule

Household membership does not mean unrestricted access to every member's information.

Access must be checked using:

- User authentication.
- Household membership.
- Role-based permissions.
- Document-level permissions.
- Consent.

- Sharing expiration.
- Resource ownership.

These checks must happen on the backend, not only in the frontend.

4. MODULE 1 — FAMILY VAULT

4.1 Overview

Family Vault is the centralized and secure document storage system of SAHAY.

Its purpose is to replace scattered physical files, mobile storage, and disconnected document folders with one organized digital family vault.

Users can upload, categorize, search, retrieve, share, and manage documents.

Family Vault stores documents. It does not independently perform all document intelligence. Document Doctor is responsible for understanding document contents.

4.2 Document Categories

A. Identity Documents

- Aadhaar Card.
- PAN Card.
- Voter ID.
- Passport.
- Driving Licence.
- Ration Card.
- Birth Certificate.
- Marriage Certificate.

B. Financial Documents

- Bank Statements.
- Loan Statements.
- Credit Card Statements.
- Insurance Policies.
- Income Tax Documents.
- Investment Statements.
- Property Tax Documents.

C. Health Documents

- Medical Prescriptions.
- Laboratory Reports.
- Hospital Discharge Summaries.
- Vaccination Records.
- Medical History.
- Health Insurance Documents.

D. Education Documents

- School ID Cards.

- Report Cards.
- Fee Receipts.
- Admit Cards.
- Mark Sheets.
- Educational Certificates.

E. Household Documents

- Electricity Bills.
- Water Bills.
- Gas Bills.
- Rent Agreements.
- Property Documents.
- Maintenance Receipts.
- Vehicle Registration.
- Vehicle Insurance.
- Product Warranties.

F. Legal and Other Documents

- Legal Agreements.
- Court-related Documents.
- Other personal records.
- Custom document categories.

4.3 Document Upload

Users should be able to upload documents through:

- Mobile camera.
- Mobile gallery.
- Desktop file picker.
- PDF upload.
- Image upload.

Supported formats for the initial implementation:

- PDF.
- JPG.
- JPEG.
- PNG.

Additional formats can be introduced later.

Upload Workflow

1. User selects a document.
2. SAHAY validates the file type and size.
3. User selects the document category or allows classification.
4. User selects the relevant household member.

5. User adds optional tags or notes.
6. The file is encrypted.
7. The encrypted file is uploaded to private object storage.
8. Document metadata is stored in PostgreSQL.
9. A Document Doctor processing job is created.
10. The document becomes available in Family Vault according to its processing and access state.

4.4 Document Management

Users should be able to:

- View uploaded documents.
- Search documents.
- Filter by category.
- Filter by household member.
- Add tags.
- Edit permitted metadata.
- Rename documents.
- Download authorized documents.
- View document details.
- View document versions.
- Delete documents.
- Export their data.

4.5 Document Search

Search should support:

- Document name.
- Document category.
- Household member.
- Tags.
- Extracted document information where available.

Example: A user searches for "Aarav school fee receipt." SAHAY should retrieve matching documents the user is authorized to access.

4.6 Document Sharing

Users may temporarily share selected documents with:

- Hospitals.
- Doctors.
- Schools.
- Insurance agents.
- Trusted family members.
- Other authorized parties.

Sharing controls:

- Selected document.
- Recipient.
- Purpose of sharing.
- Access permissions.
- Expiration time.
- Revocation.
- Access history.

SAHAY must not create unrestricted public document links.

4.7 Access Requests

A family member may request access to a private document.

Workflow:

1. Member requests access.
2. Document owner receives a request.
3. Owner reviews the request.
4. Owner approves or declines.
5. If approved, an access grant is created.
6. Access expires according to the configured rules.
7. The action is recorded in the audit log.

4.8 Document Security

Family Vault must use:

- AES-256-GCM encryption for document encryption.
- Envelope encryption.
- Per-document Data Encryption Keys (DEKs).
- Key Encryption Keys (KEKs) managed through a KMS or secure key-management service.
- Private S3-compatible object storage.
- Short-lived presigned URLs where required.
- TLS for data transmission.
- SHA-256 integrity verification.
- Malware scanning.
- Backend authorization before document access.

The application must never expose permanent public storage URLs.

4.9 Document Lifecycle

A document moves through the following lifecycle:

Upload → Validation → Encryption → Storage → Metadata Creation → AI Processing → User Verification → Normal Access → Sharing or Updating → Retention or Deletion.

4.10 Family Vault and Other Modules

Connected module	Relationship
Document Doctor	Reads and extracts document information
Sahay Daily	Receives verified deadlines and obligations
Health Shield	Uses authorized medical documents
Lakshya	Can link supporting documents to goals
AI Assistant	Retrieves authorized document information
Sahay Shield	Protects access, sharing, and storage

5. MODULE 2 — DOCUMENT DOCTOR

5.1 Overview

Document Doctor is SAHAY's document intelligence engine.

It uses OCR, document classification, information extraction, validation, and AI-assisted interpretation to convert unstructured documents into structured information.

Its primary objective is to eliminate the need for users to manually read and enter every important document detail.

5.2 Main Responsibilities

1. Document understanding.
2. Document classification.
3. Information extraction.
4. Bill detection.
5. EMI detection.
6. Insurance information extraction.
7. Payment-related information extraction.
8. Prescription information extraction.
9. Expiry date detection.
10. Recurring obligation detection.
11. Structured data generation.
12. Confidence scoring.
13. User verification.

5.3 Input Sources

A. Uploaded Documents

- PDFs.
- Scanned documents.
- Images.
- Screenshots.

B. Financial SMS Events

With explicit user permission, SAHAY may process relevant financial SMS events.

Examples:

- EMI debit confirmation.
- Bill payment confirmation.
- Insurance premium payment confirmation.

The system should extract only relevant structured events.

Where supported, parsing should happen on-device. Only minimal structured information should be sent to the backend.

SAHAY should not collect OTPs or treat all personal SMS messages as generic financial records.

C. Email Bills

With explicit permission, users may connect email accounts to identify relevant bills and statements.

Only relevant information should be processed.

D. Manual Entry

Users should be able to enter information manually when automatic extraction is unavailable or incorrect.

5.4 Document Processing Pipeline

1. File validation.
2. Image preprocessing.
3. OCR.
4. Document classification.
5. Schema-based information extraction.
6. Rule-based validation.
7. Confidence scoring.
8. PII masking where required.
9. User verification.
10. Structured record creation.
11. Storage.
12. Triggering connected workflows.

Step 1: Validation

Check:

- File type.
- File size.
- File integrity.
- Whether the document is readable.
- Whether the file contains malware.

Step 2: Preprocessing

For image-based documents:

- Image resizing.
- Noise removal.
- Deskewing.
- Contrast adjustment.
- Rotation correction.

Step 3: OCR

OCR converts text inside documents into machine-readable text.

Initial option:

- Tesseract OCR.

Possible future alternatives:

- Google Document AI.
- AWS Textract.

Step 4: Classification

The system identifies the document type.

Examples:

- Electricity Bill.
- Insurance Policy.
- Bank Statement.
- Loan Statement.
- Medical Prescription.
- Laboratory Report.
- School Fee Receipt.

Step 5: Information Extraction

The system extracts relevant fields using predefined schemas.

Example: Electricity Bill

- Electricity provider.
- Consumer number.
- Bill period.
- Bill amount.
- Due date.
- Late payment date.
- Payment status, if available.

Example: Insurance Policy

- Insurance provider.
- Policy number.
- Policy type.
- Policyholder.
- Premium amount.
- Premium due date.
- Expiry date.

Example: Medical Prescription

- Patient name.
- Doctor name.
- Medicine name.

- Dosage.
- Frequency.
- Duration.
- Prescription date.

5.5 Confidence and Validation

AI extraction is not always correct.

Therefore:

- Every extracted field should have an associated confidence score where supported.
- Important extracted information should be linked to source evidence.
- Missing information must remain null.
- The system must never invent missing values.
- Low-confidence fields should be presented for user verification.
- Users must be able to correct extracted information.

5.6 Recurring Obligation Detection

Document Doctor should identify recurring financial obligations.

Examples:

- Electricity bills.
- Water bills.
- Gas bills.
- Rent.
- EMIs.
- Insurance premiums.
- Subscriptions.
- School fees.

The system should:

1. Extract the obligation details.
2. Identify the expected recurrence.
3. Compare against existing obligations.
4. Update an existing record where appropriate.
5. Avoid duplicate obligation creation.
6. Ask the user to verify uncertain recurrence patterns.

5.7 Financial Event Matching

Example: A user receives an EMI debit SMS.

Document Doctor:

1. Identifies the message as a financial event.
2. Extracts the amount and date.
3. Searches for a matching household obligation.
4. Checks for duplicate events.

5. Evaluates whether there is enough evidence to update payment status.
6. Updates the matching record when permitted.
7. Calculates the next expected due date when sufficient information exists.

Uncertain events must be presented for confirmation rather than automatically treated as successful payments.

5.8 Integration with Sahay Daily

After verification, Document Doctor sends structured information to Sahay Daily.

Example:

Electricity bill uploaded → Due date extracted → User verifies → Obligation created → Reminder scheduled → Payment evidence → Update paid → Predict next due date.

6. MODULE 3 — SAHAY DAILY

6.1 Overview

Sahay Daily is the central household operations and planning dashboard.

It combines the household's daily activities, upcoming deadlines, recurring financial obligations, and expected cash flow.

The module has three main views:

1. Today.
2. Calendar.
3. Money.

6.2 Today View

The Today view displays household activities based on urgency and date.

Categories

Critical

Items requiring immediate attention.

Examples:

- Overdue payment.
- Urgent document expiry.
- Important missed deadline.

Due Today

Activities that must be completed today.

Examples:

- Electricity bill.
- Medicine refill.
- EMI.
- Assigned household task.

Due Soon

Activities approaching their deadlines.

Upcoming

Future household obligations.

Completed

Activities that have been completed.

Snoozed

Activities temporarily postponed by the user.

6.3 Calendar View

The Calendar view provides a monthly and date-based view of household responsibilities.

It should support:

- Monthly calendar.
- Daily activity list.
- Date-based filtering.
- Category filtering.
- Household member filtering.
- Upcoming deadlines.
- Completed activities.

Filters:

- Bills.
- EMI.
- Insurance.
- School.
- Health.
- Goals.
- Documents.
- Household tasks.
- Family member.

6.4 Money View

The Money view gives users an overview of expected household finances.

It is a planning and tracking interface, not a bank account.

Main Information

- Expected monthly income.
- Expected household obligations.
- Paid obligations.
- Overdue obligations.
- Planned savings contributions.
- Projected remaining balance.
- Category-wise breakdown.
- Cash-flow warnings.

Example

Item	Amount
Expected monthly income	■80,000
Electricity	■2,000
Rent	■15,000
EMI	■12,000
School fees	■8,000
Insurance	■3,000
Planned goal contribution	■5,000

Total planned outflow = ■45,000.

Projected remaining amount = ■35,000.

This is a projection based on available information, not a guarantee of the actual bank balance.

6.5 Obligation Management

Each obligation should contain:

- Obligation ID.
- Name.
- Category.
- Amount, where applicable.
- Due date.
- Recurrence.
- Assigned member.
- Priority.
- Payment status.
- Reminder schedule.
- Notes.
- Source document, if applicable.

Supported Obligations

- Electricity bills.
- Water bills.
- Gas bills.
- Rent.
- EMIs.
- Insurance premiums.
- Credit card bills.
- School fees.
- Subscriptions.
- Taxes.
- Medicine refills.
- Doctor visits.

- Vaccinations.
- Document renewals.
- Lakshya contributions.
- Household tasks.

6.6 Recurrence Management

The obligation engine should support:

- One-time obligations.
- Daily recurrence.
- Weekly recurrence.
- Monthly recurrence.
- Yearly recurrence.
- Custom recurrence where supported.

It should calculate the next expected due date based on the recurrence configuration.

6.7 Reminders

Users should receive reminders for:

- Upcoming bills.
- Overdue payments.
- Insurance renewals.
- Document expiry.
- Medicine refills.
- Doctor appointments.
- Goal contributions.
- Household tasks.

Reminder functionality should support:

- Scheduled notifications.
- Snoozing.
- Completion.
- Notification preferences.
- Priority-based reminders.

6.8 Payment Status

Possible payment statuses:

- Pending.
- Due.
- Paid.
- Overdue.
- Partially paid, if supported.
- Cancelled.

Payment status should only be updated through:

1. Explicit user action.
2. Reliable payment evidence.
3. Authorized integrations when introduced.

An uploaded bill alone must not be interpreted as proof that a payment has been made.

6.9 Complete Example Workflow

1. User uploads an electricity bill.
2. Family Vault stores it.
3. Document Doctor extracts the amount and due date.
4. User verifies the information.
5. Sahay Daily creates an obligation.
6. The obligation appears in Today and Calendar.
7. A reminder is scheduled.
8. User pays the bill independently.
9. User marks it as paid or provides payment evidence.
10. Sahay Daily updates the obligation.
11. The next recurring obligation is generated.

7. MODULE 4 — HEALTH SHIELD

7.1 Overview

Health Shield is the centralized family health-information and emergency-support module.

It helps families organize medical records and make important health information accessible to authorized people.

It contains two primary modes:

1. Everyday Health.
2. Emergency Mode.

Health Shield is an information-management system. It is not a medical diagnosis or treatment platform.

7.2 Everyday Health

Each household member can have an authorized health profile.

Health Profile Information

- Name.
- Age.
- Blood group.
- Height.
- Weight.
- Known allergies.
- Chronic conditions.
- Previous surgeries.
- Current medicines.
- Vaccination records.

- Medical history.
- Doctor information.
- Hospital information.
- Health insurance details.
- Emergency contacts.

Access must follow the household's privacy and permission rules.

7.3 Prescription Management

Users can upload medical prescriptions.

Workflow

1. User uploads a prescription.
2. Family Vault stores the document.
3. Document Doctor processes the document.
4. Prescription information is extracted.
5. Extracted details are displayed.
6. User verifies the information.
7. Confirmed medicines are added to the health profile.
8. Medicine duration and refill information are calculated when sufficient data is available.
9. Relevant reminders are created in Sahay Daily.

Extracted Information

- Patient name.
- Doctor name.
- Prescription date.
- Medicine name.
- Dosage.
- Frequency.
- Duration.

The system must not independently recommend medicines or alter prescribed dosage.

7.4 Laboratory Report Management

Users can upload:

- Blood test reports.
- Urine test reports.
- Other laboratory reports.
- Diagnostic reports.

Workflow

1. Upload report.
2. Store in Family Vault.
3. Extract report type and date.
4. Extract supported values.

5 Present the extracted information.

6 User verifies the information.

7 Store the confirmed information in health history.

8 Allow temporary authorized sharing with a doctor.

The system should preserve the original report.

It should not interpret a laboratory result as a definitive diagnosis.

7.5 Medicine Reminders

When a prescription contains sufficient information, SAHAY can help track medicine schedules.

Information includes:

- Medicine name.
- Dosage.
- Frequency.
- Duration.
- Start date.
- Expected end date.

Sahay Daily can display refill or medicine-related reminders.

These reminders are organizational aids, not substitutes for medical instructions.

7.6 Emergency Mode

Emergency Mode provides a quick-access health information card.

Emergency Health Card

- Name.
- Age.
- Blood group.
- Known allergies.
- Important medical conditions.
- Current medicines.
- Emergency contact numbers.
- Doctor details.
- Hospital details.
- Insurance information.
- Critical medical documents.

Emergency Sharing

The user can authorize access for selected people.

Security controls:

- Explicit permission.
- Time-limited access.
- Revocable sharing.
- Audit logs.

- Separate consent for location sharing.

Location sharing must not be enabled without appropriate permission.

7.7 Emergency Workflow

1. User activates Emergency Mode.
2. SAHAY displays the authorized emergency information.
3. User chooses whether to share information.
4. SAHAY creates a controlled access grant.
5. The recipient accesses permitted information.
6. Access expires or is revoked.
7. The activity is recorded.

8. MODULE 5 — LAKSHYA GOAL VAULTS

8.1 Overview

Lakshya is the family savings-goal management system.

It helps families plan financial goals, assign contributors, monitor contributions, and understand whether they are progressing toward their targets.

Lakshya is not a bank account, investment platform, or payment-processing system.

8.2 Goal Categories

A. Child Goals

- Education.
- Higher Education.
- Overseas Education.
- Marriage.
- Vehicle.
- Custom goals.

B. Family Goals

- Emergency Fund.
- Medical Fund.
- Home Purchase.
- Vehicle Purchase.
- Travel.
- Festival.
- Wedding.
- Custom goals.

C. Elder Care Goals

- Medical Expenses.
- Elder Care.
- Retirement Support.

D. Protection Goals

- Emergency Protection.
- Insurance Premium.
- Document Renewal.

8.3 Goal Creation

Each goal should contain:

- Goal ID.
- Goal name.
- Category.
- Beneficiary.
- Target amount.
- Target date.
- Current savings.
- Monthly contribution.
- Contributors.
- Contribution schedule.
- Contribution history.
- Withdrawal history.
- Milestones.
- Linked documents.
- Notes.
- Goal status.

8.4 Goal Creation Workflow

1. User selects a goal category.
2. User enters the goal name.
3. User selects the beneficiary.
4. User enters the target amount.
5. User enters the target date.
6. User selects contributors.
7. User configures the contribution schedule.
8. SAHAY calculates the required contribution.
9. User reviews the plan.
10. User confirms creation.
11. Lakshya creates the goal.
12. Sahay Daily schedules relevant reminders.

8.5 Contribution Types

- Monthly contribution.
- One-time contribution.

- Bonus contribution.
- Gift contribution.
- Manual contribution.
- Simulated automatic savings schedule.

In the MVP, contribution records are tracking entries. They do not represent money automatically transferred into a bank account.

8.6 Goal Progress

Progress percentage:

Progress = (Current Savings / Target Amount) × 100

Remaining amount:

Remaining Amount = Target Amount – Current Savings

Required monthly contribution:

Required Monthly Contribution = Remaining Amount / Remaining Months

If the target date has passed or the remaining period is zero, the application must handle that situation separately rather than dividing by zero.

8.7 Goal Status

Possible statuses:

- Active.
- On Track.
- At Risk.
- Behind.
- Paused.
- Completed.
- Withdrawn.
- Cancelled.

8.8 Goal Risk Detection

Lakshya should identify potential problems:

- Missed contributions.
- Reduced contributions.
- Insufficient progress.
- Deadline approaching.
- Large withdrawals.
- Projected shortfall.

8.9 Goal Suggestions

Based on the available goal data, SAHAY can suggest:

- Increase monthly contribution.
- Extend the target date.
- Reduce the target amount.

- Add a one-time contribution.

These suggestions are planning assistance, not investment or financial advice.

8.10 Goal Roles

Goal Owner

Creates and manages the goal.

Contributor

Contributes toward the goal and views permitted information.

Viewer

Can view authorized progress information.

Beneficiary

The person for whom the goal is created.

8.11 Lakshya and Sahay Daily

Lakshya communicates with Sahay Daily.

Examples:

- Upcoming contribution reminders.
- Missed contribution alerts.
- Goal deadline reminders.
- Goal progress updates.

8.12 Future Payment Integration

Future versions may support authorized UPI AutoPay through an appropriate payment partner.

This is not part of the MVP.

The MVP must not initiate real bank debits or claim that simulated contributions represent actual transfers.

9. MODULE 6 — SAHAY AI ASSISTANT

9.1 Overview

Sahay AI Assistant is the conversational intelligence layer that connects the other six functional areas of SAHAY.

Users should be able to ask questions in natural language instead of manually navigating every module.

The assistant should understand the user's intent, retrieve authorized household information, and provide grounded answers.

It may also propose actions that require user confirmation.

9.2 Example Questions

Family Vault

- Where is my PAN card?
- Show my father's insurance document.
- When does my passport expire?
- Find the school fee receipt.

Document Doctor

- Summarize this insurance policy.
- What is the due date on my electricity bill?
- What information is present in this loan statement?

Sahay Daily

- What payments are due this week?
- Which bills are overdue?
- How much are our expected household expenses this month?
- What tasks are assigned to me?

Health Shield

- What is my blood group?
- Which medicines are recorded in my profile?
- Show the latest uploaded laboratory report.
- What emergency information is available for an authorized family member?

Lakshya

- What is the progress of our education goal?
- How much is remaining?
- What contribution is due next month?
- Which goals are currently at risk?

Household Management

- Summarize the family's upcoming responsibilities.
- What needs attention this week?
- Create a household task for tomorrow.

9.3 AI Architecture

The AI Assistant should use permission-aware Retrieval-Augmented Generation (RAG).

The system should not provide the LLM with unrestricted access to the database or document storage.

Processing Workflow

1. User submits a question.
2. Language detection.
3. Intent classification.
4. User authentication.
5. Household membership validation.
6. Role and permission verification.
7. Resource-level access checks.
8. Consent and sharing-expiration checks.
9. Retrieval of authorized information.
10. PII masking where required.
11. Context preparation.
12. LLM response generation.

13 Source-grounding and response validation.

14 Answer returned to the user.

15 If an action is requested, an action proposal is created.

16 User reviews and confirms.

17 Backend executes the authorized action.

18 Audit log is generated.

9.4 RAG Data Sources

The AI Assistant may retrieve information from:

- Family Vault.
- Document Doctor.
- Sahay Daily.
- Health Shield.
- Lakshya.
- Household membership and permission records.

The assistant should retrieve only information the current user is authorized to access.

9.5 Document Question Answering

Example:

User asks: "When is my health insurance due?"

The system:

1 Identifies the intent.

2 Checks access permissions.

3 Searches authorized insurance documents and records.

4 Retrieves the relevant policy information.

5 Generates an answer using the available source.

6 Shows the source document or record when appropriate.

If the information is unavailable, the assistant must say so.

9.6 Action-Based AI

The assistant can propose supported actions such as:

- Create an obligation.
- Create a savings goal.
- Add a household task.
- Create a reminder.
- Update an authorized record.

Sensitive actions require confirmation.

The AI model must not directly modify database records.

Instead:

AI → Backend action proposal → Permission validation → User confirmation → Backend tool execution → Audit log.

9.7 AI Security Rules

1. No unrestricted database access for the LLM.
2. No unrestricted object-storage access.
3. No unauthorized document retrieval.
4. No automatic sharing of private documents.
5. No automatic movement of money.
6. No medical diagnosis or prescription generation.
7. No fabricated document information.
8. No exposure of unnecessary personal information.
9. No public AI-model training using private household data.
10. Document contents must be treated as untrusted input.

9.8 Language Support

The system should be designed to support English and additional Indian languages.

Language detection, multilingual embeddings, and appropriate LLM capabilities can be used.

Indic speech recognition and text-to-speech may be added as future capabilities.

10. MODULE 7 — SAHAY SHIELD

10.1 Overview

Sahay Shield is the cross-cutting security, privacy, and trust infrastructure of SAHAY.

It protects information and interactions across every module.

It is not a separate scam detection product or an additional household management dashboard.

Its purpose is to ensure that household data is accessed, processed, shared, and stored safely.

10.2 Main Responsibilities

- Authentication.
- Session security.
- Household isolation.
- Role-based access control.
- Document-level access control.
- Consent management.
- Encryption.
- Key management.
- Secure document sharing.
- PII masking.
- Audit logging.
- Security event monitoring.
- Data retention.
- Data export.
- Data deletion.

- Breach notification.
- Security settings.

10.3 Authentication

The authentication system should support:

- User registration.
- Secure login.
- Password hashing.
- Session management.
- Token-based authentication.
- Refresh-token rotation.
- Session revocation.
- Device/session management.
- Rate limiting.
- Optional OTP verification.
- Optional biometric application lock.

Recommended password hashing:

- Argon2id.

Access tokens should be short-lived.

Refresh tokens should be rotated and revocable.

10.4 Authorization

SAHAY should use multiple authorization layers.

Layer 1: Authentication

Is the user logged in?

Layer 2: Household Authorization

Does the user belong to the requested household?

Layer 3: Role Authorization

Does the user's household role permit the requested operation?

Layer 4: Resource Authorization

Does the user have permission to access this specific document or record?

Layer 5: Consent and Expiration

Is the access grant valid and unexpired?

Every sensitive operation should be checked by the backend.

10.5 Encryption

Data in Transit

Use TLS 1.3 where supported.

Data at Rest

Use encryption for stored documents, backups, and sensitive data.

Document Encryption

Use:

- AES-256-GCM.
- Unique per-document DEKs.
- KMS-managed KEKs.
- Envelope encryption.
- Secure key rotation.

The key-management service should protect the encryption keys separately from the stored documents.

10.6 Consent Management

Users must be able to manage permissions for:

- Document sharing.
- Health information access.
- Location sharing.
- Connected email access.
- Financial SMS processing.
- Other sensitive data processing.

Consent records should include:

- User.
- Purpose.
- Scope.
- Granted permissions.
- Timestamp.
- Expiration.
- Revocation status.

10.7 Audit Logging

Important activities should be recorded.

Examples:

- Login.
- Failed login.
- Document upload.
- Document view.
- Document download.
- Document sharing.
- Access request.
- Access approval.
- Access revocation.
- Health information access.
- AI retrieval of sensitive information.

- AI action execution.
- Permission changes.
- Data export.
- Account deletion.

Audit records should identify the actor, action, resource, timestamp, and relevant outcome.

Logs must not unnecessarily expose sensitive document contents.

10.8 Data Lifecycle

Sahay Shield governs the full data lifecycle:

Collection → Validation → Encryption → Storage → Access Control → Processing → Sharing → Retention → Export or Deletion.

10.9 Privacy Principles

- Privacy by default.
- Least privilege.
- Data minimization.
- Purpose limitation.
- Explicit consent.
- Encryption.
- Controlled sharing.
- Access revocation.
- Auditability.
- User control.
- No public document links.
- No public model training on private household information.

11. HOW THE SEVEN MODULES WORK TOGETHER

The main strength of SAHAY is the connection between its modules.

11.1 Electricity Bill Workflow

1. User uploads an electricity bill.
2. Family Vault securely stores the document.
3. Document Doctor identifies the bill.
4. OCR extracts the provider, amount, and due date.
5. User verifies the extracted information.
6. Document Doctor sends verified data to Sahay Daily.
7. Sahay Daily creates an obligation.
8. The obligation appears in Today and Calendar.
9. A reminder is scheduled.
10. User confirms payment.
11. Sahay Daily updates payment status.

12 The AI Assistant can answer questions about the bill.

13 Sahay Shield records sensitive access and actions.

11.2 Medical Prescription Workflow

1 User uploads a prescription.

2 Family Vault stores it.

3 Document Doctor extracts the prescription information.

4 User verifies the extracted fields.

5 Health Shield stores the confirmed medicine information.

6 Sahay Daily creates relevant reminders when sufficient information exists.

7 The AI Assistant answers authorized questions.

8 Sahay Shield protects health data and access.

11.3 Savings Goal Workflow

1 User creates an education savings goal.

2 Lakshya stores the goal details.

3 Contributors are assigned.

4 Required contributions are calculated.

5 Sahay Daily schedules contribution reminders.

6 Users record contributions.

7 Lakshya updates progress.

8 Risk indicators are calculated.

9 The AI Assistant summarizes progress.

10 Sahay Shield protects goal information.

11.4 Emergency Workflow

1 User maintains a health profile.

2 Medical documents are stored in Family Vault.

3 Health Shield maintains the emergency health card.

4 User authorizes an emergency share.

5 Selected information becomes accessible for the permitted period.

6 Sahay Shield records access and manages expiration or revocation.

12. TECHNICAL STACK

The following is the proposed technology stack for the SAHAY implementation.

12.1 Frontend

Technology	Purpose
React	User interface
TypeScript	Type safety
Vite	Development and build tooling

Technology	Purpose
Tailwind CSS	Styling
shadcn/ui	Reusable UI components
React Router	Navigation
TanStack Query	Server-state management
Axios	HTTP communication
Recharts	Charts and visualizations
Lucide	Icons
Web Speech API	Browser speech capabilities where supported

The frontend should be modular and feature-oriented.

12.2 Backend

Technology	Purpose
Python 3.12	Backend programming language
FastAPI	REST API framework
SQLAlchemy 2.0	Database ORM
PostgreSQL	Main relational database
pgvector	Vector storage
Alembic	Database migrations
Pydantic v2	Validation and schemas
Redis	Caching and job coordination
Celery	Background task processing
Pytest	Testing

12.3 AI and Document Processing

Technology	Purpose
Tesseract OCR	Text extraction
Google Document AI	Optional advanced document processing
AWS Textract	Optional document extraction alternative
LLM API	Language understanding and generation
Sentence Transformers	Embeddings
pgvector	Semantic retrieval
PostgreSQL full-text search	Keyword retrieval
Presidio	PII detection and masking
Indic ASR/TTS	Potential future Indian-language speech support

Not every optional service needs to be included in the first MVP.

12.4 Infrastructure

Technology	Purpose
Docker	Application containerization
Docker Compose	Local service orchestration
MinIO	Local S3-compatible object storage
AWS S3	Potential production document storage
AWS KMS / Vault	Key management
Firebase Cloud Messaging	Push notifications
Git	Version control
GitHub	Repository hosting
GitHub Actions	CI/CD
Prometheus	Metrics
Grafana	Monitoring dashboards
Sentry	Error tracking

12.5 Why This Stack?

The proposed stack provides:

- A modular frontend.
- A Python backend suitable for AI and ML integration.
- Relational data storage.
- Semantic retrieval capabilities.
- Asynchronous document processing.
- Secure file storage.
- Containerized development.
- Monitoring and deployment support.

The stack is a proposed engineering design. Individual tools may be changed if a technical requirement justifies the change.

13. SYSTEM ARCHITECTURE

13.1 High-Level Architecture

The application follows a modular client-server architecture.

```
SAHAY USERS
|
REACT + TYPESCRIPT
FRONTEND
|
HTTPS / REST
|
FASTAPI BACKEND
|
AUTHENTICATION & SECURITY
SAHAY SHIELD
|
MODULAR APPLICATION
|
+--------------+--------------+
| | |
v v v
FAMILY VAULT DOCUMENT DOCTOR SAHAY DAILY
| | |
+--------------+--------------+
```

```
|
+--------------+--------------+
| | |
v v v
HEALTH SHIELD LAKSHYA AI ASSISTANT
| | |
+--------------+--------------+
|
DATA SERVICES
|
+---------------+---------------+
| | |
v v v
POSTGRESQL REDIS/CELERY OBJECT STORAGE
+ PGVECTOR BACKGROUND MINIO / S3
PROCESSING
```

13.2 Request Flow

1. The user interacts with the React frontend.
2. The frontend sends a request to FastAPI over HTTPS.
3. Authentication is validated.
4. Sahay Shield checks permissions and consent.
5. The relevant module service processes the request.
6. The backend accesses the required database or storage service.
7. Background tasks are scheduled when necessary.
8. The response is returned to the frontend.
9. Sensitive actions are recorded in audit logs.

13.3 AI Request Flow

The AI Assistant should use backend-controlled retrieval.

```
User Question
|
Authentication
|
Intent Classification
|
Permission and Consent Checks
|
Authorized Data Retrieval
|
PII Masking
|
Context Preparation
|
LLM
|
Grounded Response
|
User
```

For an action:

```
User Request
|
Action Proposal
|
Permission Validation
|
User Confirmation
|
Backend Tool Execution
|
Audit Log
```

The LLM must never directly connect to PostgreSQL, private file storage, or unrestricted internal services.

14. DATABASE DESIGN

The database should use PostgreSQL as the primary relational database.

pgvector can support semantic search for document chunks and other authorized information.

The following is a proposed initial table structure.

14.1 Core Tables

Table	Purpose
users	User accounts
households	Household information
household_members	User-to-household membership
sessions	Active sessions
invitations	Household invitations

14.2 Family Vault Tables

Table	Purpose
documents	Document metadata
document_versions	Document version history
document_requests	Access requests
document_access_grants	Authorized sharing permissions
document_chunks	Extracted and indexed document chunks

14.3 Document Doctor Tables

Table	Purpose
document_processing_jobs	Processing jobs
extracted_document_fields	Extracted field information
financial_events	Structured financial events
processing_results	Processing output and confidence information

14.4 Sahay Daily Tables

Table	Purpose
obligations	Household obligations
obligation_events	Obligation activity history
reminders	Scheduled reminders
notification_preferences	User notification settings

14.5 Health Shield Tables

Table	Purpose
health_profiles	Member health profiles

Table	Purpose
medical_conditions	Medical condition records
allergies	Allergy records
medicines	Medicine records
vaccinations	Vaccination records
health_reports	Health report metadata
emergency_contacts	Emergency contacts
emergency_shares	Emergency access grants

14.6 Lakshya Tables

Table	Purpose
goal_vaults	Savings goals
goal_contributors	Goal participants
goal_contributions	Contribution history
goal_withdrawals	Withdrawal history
goal_milestones	Goal milestones

14.7 AI Assistant Tables

Table	Purpose
ai_conversations	Conversation records
ai_messages	Conversation messages
ai_retrieval_logs	Retrieval audit information
ai_action_requests	Proposed and confirmed actions

14.8 Sahay Shield Tables

Table	Purpose
consents	Consent records
audit_logs	General audit history
document_access_logs	Document access history
security_events	Security events
data_export_requests	Export requests
deletion_requests	Deletion requests

14.9 Database Design Rules

- Use primary keys.
- Use foreign keys.
- Add appropriate indexes.
- Add timestamps to important records.

- Enforce household isolation.
- Avoid storing unnecessary sensitive data.
- Store document metadata separately from encrypted file contents.
- Use migrations for schema changes.
- Maintain audit history for sensitive operations.
- Define deletion and retention rules.
- Use transactions for related database changes.

The proposed tables are a starting point. Final schemas should be defined with detailed field types, constraints, relationships, and indexes during implementation.

15. API DESIGN

The backend should expose versioned REST APIs.

A proposed API structure:

```
/api/v1/auth
/api/v1/users
/api/v1/households
/api/v1/family-vault
/api/v1/document-doctor
/api/v1/sahay-daily
/api/v1/health-shield
/api/v1/lakshya
/api/v1/ai-assistant
/api/v1/security
/api/v1/notifications
```

15.1 Example API Operations

Authentication

- Register.
- Login.
- Logout.
- Refresh token.
- Revoke session.

Household

- Create household.
- Invite member.
- Accept invitation.
- Update role.
- Remove member.

Family Vault

- Upload document.
- List documents.
- Search documents.
- Retrieve document.
- Update metadata.
- Request access.

- Approve access.
- Revoke sharing.
- Delete document.

Document Doctor

- Start document processing.
- Retrieve processing status.
- View extracted fields.
- Verify extracted fields.
- Correct extracted information.

Sahay Daily

- Create obligation.
- Retrieve obligations.
- Update obligation.
- Mark paid.
- Snooze reminder.
- Complete task.
- Retrieve calendar information.
- Retrieve money summary.

Health Shield

- Create health profile.
- Update profile.
- Upload health report.
- Retrieve medical history.
- Create emergency share.
- Revoke emergency share.

Lakshya

- Create goal.
- Update goal.
- Add contributor.
- Record contribution.
- Record withdrawal.
- Retrieve progress.
- Retrieve milestones.

AI Assistant

- Submit question.
- Retrieve conversation.
- Propose action.
- Confirm action.
- Cancel action.

Sahay Shield

- Manage consent.
- Retrieve authorized audit history.
- Manage access grants.
- Request data export.
- Request account deletion.

Every API must enforce authentication and authorization where applicable.

16. PROJECT FOLDER STRUCTURE

A proposed modular repository structure:

```
SAHAY/
■■■ frontend/
■ ■■■ public/
■ ■■■ src/
■ ■■■ components/
■ ■■■ layouts/
■ ■■■ pages/
■ ■■■ hooks/
■ ■■■ services/
■ ■■■ types/
■ ■■■ utils/
■ ■■■ features/
■ ■■■ family-vault/
■ ■■■ document-doctor/
■ ■■■ sahay-daily/
■ ■■■ health-shield/
■ ■■■ lakshya/
■ ■■■ ai-assistant/
■ ■■■ security/
■■■ backend/
■ ■■■ app/
■ ■ ■■■ api/v1/
■ ■ ■■■ core/
■ ■ ■■■ db/
■ ■ ■■■ models/
■ ■ ■■■ schemas/
■ ■ ■■■ services/
■ ■ ■■■ workers/
■ ■ ■■■ utils/
■ ■ ■■■ main.py
■ ■■■ alembic/
■ ■■■ tests/
■■■ docs/
■ ■■■ architecture/
■ ■■■ api/
■ ■■■ database/
■ ■■■ development/
■■■ docker-compose.yml
■■■ .env.example
■■■ .gitignore
■■■ README.md
```

The project should remain modular so each feature can be developed and tested independently.

17. MVP — MINIMUM VIABLE PRODUCT

17.1 What is the MVP?

The MVP is the first usable version of SAHAY that demonstrates the main product value through the seven connected modules.

The MVP should focus on essential functionality rather than attempting to implement every possible future integration.

17.2 MVP Scope by Module

Family Vault

- User document upload.
- PDF and image support.
- Document categorization.
- Member association.
- Document listing.
- Search.
- Secure retrieval.
- Basic controlled sharing.
- Basic audit history.

Document Doctor

- OCR.
- Basic document classification.
- Structured information extraction.
- Bill and insurance field extraction.
- Expiry detection.
- User verification.
- Simulated financial-event processing.

Sahay Daily

- Today view.
- Calendar view.
- Money view.
- Obligation creation.
- Recurring obligations.
- Payment-status tracking.
- Reminder scheduling.
- Basic household activity management.

Health Shield

- Health profiles.
- Prescription upload.
- Medical report upload.
- Basic prescription extraction.
- Medicine information tracking.
- Reminder integration.
- Emergency health card.
- Controlled emergency sharing.

Lakshya

- Goal creation.
- Goal categories.
- Beneficiary selection.
- Contributors.
- Manual contribution records.
- Progress calculation.
- Risk indicators.
- Reminders.

Sahay AI Assistant

- Conversational interface.
- Permission-aware retrieval.
- Document question answering.
- Household obligation questions.
- Goal progress questions.
- Basic action proposals.
- User-confirmed actions.

Sahay Shield

- Authentication.
- Household permissions.
- Document access control.
- Encryption.
- Consent records.
- Secure sharing.
- Audit logging.
- Basic security settings.

17.3 MVP Integrations

The initial version can use:

- PostgreSQL.
- Local MinIO.
- Tesseract OCR.
- An LLM API.
- Simulated SMS and payment confirmation events.
- Local background processing.
- Mock or basic notifications.

This keeps the initial implementation manageable.

17.4 What Is Not Included in the MVP?

- Real bank account integration.
- Automatic real UPI payments.

- Automatic bank debits.
- UPI AutoPay.
- Guaranteed government scheme eligibility.
- Medical diagnosis.
- Prescription generation.
- Investment advice.
- Tax advice.
- Unrestricted family document access.
- Public sharing of private documents.
- Training public AI models using private household information.
- Scam Shield or scam detection as a product feature.

These exclusions are intentional.

18. DEVELOPMENT PHASES

The development sequence below organizes the implementation work. It does not mean security should be postponed until the final phase.

Phase 0 — Planning and Environment Setup

- Finalize requirements.
- Finalize user journeys.
- Finalize architecture.
- Define database entities.
- Create GitHub repository.
- Configure development environments.
- Set up frontend and backend projects.
- Configure Docker.

Phase 1 — Application Foundation

- React application.
- FastAPI application.
- PostgreSQL connection.
- Database migrations.
- API structure.
- Shared UI components.
- Error handling.
- Logging.

Phase 2 — Authentication and Household Infrastructure

- User registration.
- Login.
- Password security.

- Session management.
- Household creation.
- Invitations.
- Membership.
- Roles.
- Authorization.

Secure storage and access-control foundations must be included from this phase onward.

Phase 3 — Family Vault and Document Doctor

- Document upload.
- Encrypted storage.
- Metadata management.
- Document listing.
- OCR.
- Document classification.
- Information extraction.
- Verification interface.
- Expiry detection.
- Processing jobs.

Phase 4 — Sahay Daily

- Today dashboard.
- Calendar.
- Money view.
- Obligation engine.
- Recurrence.
- Reminder scheduling.
- Payment-status management.

Phase 5 — Health Shield

- Health profiles.
- Prescription processing.
- Medical report management.
- Medicine tracking.
- Emergency health card.
- Controlled emergency sharing.

Phase 6 — Lakshya Goal Vaults

- Goal creation.
- Goal categories.
- Contributors.

- Contribution records.
- Progress calculation.
- Risk indicators.
- Reminders.

Phase 7 — Sahay AI Assistant

- AI conversation interface.
- Intent classification.
- Permission-aware retrieval.
- Document retrieval.
- RAG.
- Grounded answers.
- Action proposals.
- Confirmation workflow.

Phase 8 — Sahay Shield Hardening and Integration

- Security review.
- Encryption verification.
- Key management.
- Consent management.
- Audit logging.
- Session revocation.
- Access-control testing.
- Data lifecycle controls.
- Monitoring.
- Security dashboard.

Security foundations must already exist in earlier phases. This phase focuses on completing and hardening the security system.

Phase 9 — Testing, Deployment and Demonstration

- Unit testing.
- Integration testing.
- API testing.
- Frontend testing.
- Authorization testing.
- Security testing.
- Performance testing.
- User acceptance testing.
- Deployment.
- Documentation.
- Final product demonstration.

19. TESTING STRATEGY

Testing is essential because SAHAY handles private family information.

19.1 Unit Testing

Test individual functions:

- Document validation.
- Extraction rules.
- Due-date calculation.
- Recurrence calculation.
- Goal progress.
- Contribution calculations.
- Permission checks.

19.2 Integration Testing

Test connections between modules:

- Family Vault to Document Doctor.
- Document Doctor to Sahay Daily.
- Document Doctor to Health Shield.
- Lakshya to Sahay Daily.
- AI Assistant to authorized retrieval services.
- Sahay Shield to every protected module.

19.3 Security Testing

Test:

- Unauthorized document access.
- Cross-household access.
- Invalid sharing links.
- Expired access grants.
- Revoked permissions.
- Session expiration.
- Token rotation.
- Role changes.
- Sensitive data exposure.
- AI retrieval permission bypass.

19.4 AI Testing

Test:

- Incorrect document extraction.
- Missing information.
- Unsupported questions.
- Hallucinations.

- Unauthorized retrieval.
- Prompt injection through document contents.
- Incorrect action proposals.
- Incorrect source attribution.

19.5 User Experience Testing

Test whether:

- New users can create households.
- Users can upload documents easily.
- Users can find important information.
- Reminders are understandable.
- Health information is easy to access.
- Goal creation is straightforward.
- Elder users can navigate important screens.

20. SAMPLE DEMONSTRATION HOUSEHOLD

Use fictional information for demonstrations and testing.

20.1 Household

Household Name: Sharma Family

Member	Age	Role
Rajesh Sharma	45	Household Owner
Sunita Sharma	41	Adult Member
Aarav Sharma	14	Child
Kamla Devi	70	Elder Member

All names and information are fictional demonstration data.

20.2 Sample Documents

- Rajesh's PAN Card.
- Sunita's Aadhaar Card.
- Aarav's School Fee Receipt.
- Family Electricity Bill.
- Rajesh's Insurance Policy.
- Kamla Devi's Medical Prescription.
- Family Health Insurance Document.

20.3 Sample Obligations

- Electricity Bill.
- Water Bill.
- Monthly Rent.
- Education Fees.

- Insurance Premium.
- EMI.
- Medicine Refill.

20.4 Sample Lakshya Goals

- Aarav's Higher Education.
- Family Emergency Fund.
- Kamla Devi's Medical Fund.

The demonstration must never use real private documents or actual sensitive information.

21. SUCCESS CRITERIA

The MVP should be considered functionally successful when it can demonstrate the following:

1. A user can register and log in.
2. A household can be created.
3. Members can be invited.
4. Roles and permissions are enforced.
5. Documents can be uploaded securely.
6. Documents can be categorized and searched.
7. Document Doctor can extract supported information.
8. Users can verify extracted fields.
9. Verified obligations can be created.
10. Sahay Daily can display obligations.
11. Reminders can be scheduled.
12. Payment status can be updated through an appropriate confirmation.
13. Health profiles can be maintained.
14. Prescriptions and reports can be organized.
15. Emergency information can be shared through controlled access.
16. Lakshya goals can be created.
17. Contributions can be recorded.
18. Goal progress can be calculated.
19. The AI Assistant can answer questions using authorized information.
20. AI actions require confirmation.
21. Sensitive actions are recorded.
22. Users cannot access unauthorized household information.

22. BUSINESS MODEL

SAHAY may use a freemium subscription model.

The following is a proposed business model, not a finalized pricing decision.

22.1 Free Plan

Possible features:

- Limited document storage.
- Basic document organization.
- Basic reminders.
- Emergency health card.
- Limited household members.
- One goal vault.
- Basic AI usage limits.

An initial proposal could include up to 20 documents and two household members.

22.2 Premium Plan

Possible features:

- Higher or unlimited document storage within fair-use limits.
- More household members.
- Advanced document intelligence.
- Advanced reminders.
- Extended health and financial tracking.
- Multiple Lakshya goals.
- Financial projections.
- Advanced AI usage.
- Enhanced security dashboard.

22.3 Future Partnerships

Potential partnerships may include:

- Healthcare providers.
- Insurance providers.
- Educational institutions.
- Document service providers.
- Payment partners.

Partnerships must not compromise user privacy.

22.4 Data Monetization Principle

SAHAY should not sell private household documents or personal household information.

User trust and privacy should be central to the business model.

23. FUTURE EXPANSION POSSIBILITIES

These are possible future directions, not commitments for the MVP.

- More Indian language support.
- Voice-based household assistant.
- Advanced document understanding.

- More sophisticated financial forecasting.
- Authorized banking integrations.
- Authorized UPI AutoPay through partners.
- Advanced family analytics.
- More sophisticated notification channels.
- Additional accessibility features.
- Advanced household automation.
- Additional integrations with external services.

Any future feature must follow SAHAY's privacy, permission, and security principles.

24. DEVELOPMENT RULES FOR AI CODING ASSISTANTS

Any AI coding assistant working on SAHAY must follow these instructions.

24.1 Understand Before Coding

1. Read this complete document before starting implementation.
2. Inspect the existing repository.
3. Understand the current architecture.
4. Identify existing modules and reusable components.
5. Do not assume the repository is empty.

24.2 Follow the Architecture

- Use React and TypeScript for the frontend.
- Use FastAPI and Python for the backend.
- Use PostgreSQL for relational data.
- Use pgvector for vector retrieval where required.
- Use Redis and Celery for background processing where appropriate.
- Follow the modular folder structure.
- Keep business logic out of frontend components.
- Use Pydantic schemas for backend validation.
- Use database migrations.

24.3 Implement Only the Requested Scope

- Follow the defined development phase.
- Do not implement future features prematurely.
- Do not add unapproved modules.
- Do not introduce Scam Shield.
- Do not add Scheme Finder as a core module without a product decision.
- Do not remove existing functionality without approval.

24.4 Security Rules

- Never hardcode secrets.

- Use environment variables.
- Never bypass backend authorization.
- Never expose private storage.
- Never give the LLM unrestricted database access.
- Never allow automatic unauthorized document sharing.
- Never initiate real payments in the MVP.
- Use synthetic demonstration data.
- Enforce household-level data isolation.

24.5 Code Quality

- Write modular code.
- Use meaningful names.
- Add error handling.
- Add appropriate tests.
- Document important functions.
- Avoid unnecessary dependencies.
- Avoid duplicated business logic.
- Maintain API documentation.
- Keep frontend and backend types consistent.

24.6 Architecture Changes

Before making a significant architectural change:

1. Explain the proposed change.
2. Explain why it is required.
3. Identify affected modules.
4. Describe the trade-offs.
5. Obtain approval.

Do not silently change product requirements or architecture.

25. IMPORTANT PRODUCT BOUNDARIES

The following distinctions must remain clear throughout development.

Component	Responsibility
Family Vault	Stores and manages documents
Document Doctor	Understands documents and extracts information
Sahay Daily	Manages household responsibilities and financial planning
Health Shield	Organizes health records and emergency information
Lakshya	Tracks family savings goals
Sahay AI Assistant	Provides conversational intelligence and confirmed actions
Sahay Shield	Protects security, permissions, privacy, and trust

25.1 Important Rules

- Family Vault is the document storage layer.
- Document Doctor is the document intelligence layer.
- Sahay Daily is the household operations layer.
- Health Shield is the health information layer.
- Lakshya is the savings-goal tracking layer.
- Sahay AI Assistant is the conversational intelligence layer.
- Sahay Shield is the security and privacy layer.

These responsibilities should not become unnecessarily mixed.

26. FINAL PRODUCT DEFINITION

SAHAY is a unified AI-powered household life-management platform that brings family documents, document intelligence, everyday responsibilities, health information, savings goals, and conversational AI into one secure ecosystem.

It is designed to help families move from fragmented household management to organized, intelligent, and permission-controlled management.

Its seven core modules work together:

1. **Family Vault** keeps important documents organized and secure.
2. **Document Doctor** understands documents and converts information into structured records.
3. **Sahay Daily** manages household responsibilities, deadlines, reminders, and financial planning.
4. **Health Shield** organizes family health information and supports controlled emergency access.
5. **Lakshya Goal Vaults** help families plan and track savings goals.
6. **Sahay AI Assistant** connects the modules through natural-language interaction and confirmed actions.
7. **Sahay Shield** protects the entire ecosystem through security, privacy, permissions, and auditing.

The central idea is simple:

SAHAY should help a family know what it owns, understand what it needs to do, remember what is important, plan for what is ahead, and securely access the information it needs.

End of SAHAY Master Product Documentation

This document is the central product and technical reference. Detailed implementation specifications, API schemas, database field definitions, UI wireframes, and deployment configurations should be maintained as separate supporting documents as development progresses.